Making Economic Decisions: Opportunity Cost, Rents, and Incentives

ECON 201: Principles of Microeconomics
Lecture 3

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Where We Left Off

- Capitalism raised what one worker could produce in two ways:
  - **incentives** for firms to adopt new technology.
  - **specialization** on a new scale.
- Both words carry a lot of weight, so this week we unpack them.

Today: *opportunity cost*, *economic rent*, and *incentives*, which explain how we choose.

Next class: *comparative advantage* and *specialization*, which decide who does what.

Everything Starts With a Decision

- What happens in the economy is the result of decisions by individuals, firms, and governments, so explaining the economy means explaining **choices**.
- Economists explain choices by comparing two things:
  - the **cost**: what you have to pay or put in to take the action.
  - the **benefit**: what the action is worth to you.
- Some costs and benefits come in dollars already. Others you have to put a number on yourself.
  - **E.g.** your parking permit has a price. The half hour you spend circling the structure for a spot does not, yet it clearly costs you: time, patience, peace of mind.

How Do You Put a Number on That?

How do you put a number on the cost of circling around the parking lot?

- Imagine you could pay for a reserved spot and skip looking for one each morning. How much more would you pay per month? \$5? \$20? \$50?
- At some price you stop and say: no thanks, I will just drive around and find a spot.
- That price measures the **cost** of circling: **the most you would pay to *avoid it***.

Attaching a monetary value to a **benefit** works **the same way**: what something is worth to you is **the most you would pay to *get it***.

First, a Question

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A good friend invites you to Saturday's Dodgers game. **What is the most you would pay** for the whole day, including ticket, parking, food, and gas?

Answers are anonymous. One number, in dollars.

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That Number Has a Name

The most you would pay is your **willingness to pay**, and it is your **benefit**: what the day is worth to you.

- It is not what the day costs. It is what you would give up to have it.
- Everyone wrote a different number, which makes sense: your personal value differs because of how much you like baseball, how introverted or extroverted you are, etc.

Would You Go?

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Ticket, parking, food, and gas come to **$100** for the day. **Do you go?**

Yes, no, or I am undecided.

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The Rule So Far

One way to make this decision would be to look at the **net benefit**:

$$\text{net benefit} = \text{benefit} - \text{direct cost}$$

So a simple rule could be:

- Net benefit $\geq 0 \rightarrow$ Go to the game!
- Net benefit $< 0 \rightarrow$ Stay home!

However, something is missing from this equation: what else could you have done with your Saturday? Whether the game is worth it depends on your **next best alternative**, sometimes also called your **outside option**.

What Else Could Saturday Have Been?

- The campus rec center needs someone for a four-hour shift on Saturday. It pays **$30 an hour**, so **$120** for the afternoon.
- Take the shift and you miss the game. Go to the game and you miss the $120.
- The $120 is the **benefit** of the shift, and it is already in dollars. What does the shift **cost** you?
  - Four hours of your Saturday at a front desk. That is a cost you have to put a number on yourself.

One More Question

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Forget the game for a second. **What is the least they would have to pay you** to work a four-hour shift on a Saturday, if your other option were a free afternoon at home?

One number, in dollars.

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The Net Benefit of the Shift

- Your poll answer is what giving up four hours of your Saturday costs you.
- The shift pays \$120, so your net benefit from working is \$120 minus your own number.
- If you asked for more than \$120, you would turn the shift down even with no game on.

Notice the switch. For the game the **cost** came priced and you judged the **benefit**. For the shift the **benefit** comes priced and you judged the **cost**.

Would You Still Go?

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The rec center shift is on the table: $120 for four hours. **Do you still go to the game?**

Yes, no, or I am undecided.

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How Would an Economist Decide?

Meet Sam, who is an econ major and also a sports lover.

- The game is worth **$200** to Sam: the most Sam would pay for the day.
- You would have to pay Sam at least **$80** to work a shift on a Saturday: that is the cost of the shift to Sam.

	Go to the game	Work the shift
Benefit	**$200**	$120
Direct cost	$100	**$80**
Net benefit	**$100**	**$40**

Sam should go to the game: it brings more net benefit than the outside option of working the shift.

Opportunity Cost

- Often, choosing one alternative means giving up another, and what you give up is a real cost you have to consider. This is a key idea in economics.
- Formally, **opportunity cost** is what you give up when you take one action instead of the next best alternative, your **outside option**.
- Sam's opportunity cost of going to the game is \$40: the net benefit of the outside option, the shift.

The final decision rule: take the action if its **net benefit** is greater than its **opportunity cost**. For Sam: \$100 > \$40, so go to the game.

Some Other Useful Terms

- **Economic cost**:

$$\text{economic cost} = \text{direct cost} + \text{opportunity cost}$$

  For the game: \$100 + \$40 = \$140. (The decision rule can also be stated as benefit > economic cost: \$200 > \$140.)

- **Economic rent**:

$$\text{economic rent} = \text{net benefit of the action chosen} - \text{opportunity cost}$$

  For Sam: \$100 - \$40 = \$60. Positive rent means the move is worth making; zero rent means you are indifferent.

How Would You Decide?

Worksheet, Activity 1: Sam's table, with **your own two numbers** in it.

Fill it in, make the call, and check it against what you told the polls.

Quick Check

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You have $10,000 in savings. A friend asks you to invest it in their business, and offers to pay you **$15 every month**, guaranteed, for as long as they hold it. **Do you take the deal?**

Yes, no, or I am undecided.

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What Is the Opportunity Cost of Taking the Deal?

- $15 a month, guaranteed, from someone you trust, sounds like easy money. But your savings had an **outside option**.
- A safe savings account paying 2% would earn about **$17 a month** on $10,000.

The opportunity cost of the deal is $17 a month, more than the $15 it pays. A deal that pays you every month can still lose you money.

Opportunity Costs You Cannot See

Sometimes what you give up never shows up as a payment, so it is easy to miss.

- **A degree.** The biggest cost is not tuition but the earnings given up while studying. Enrollment rises in recessions, when those forgone earnings are low, and more students drop out when jobs are plentiful.
- **A paid-off house.** The owner pays no rent, but gives up the rent the house could earn. That forgone rent matters enough that GDP statisticians estimate it and count it in GDP.

Right Numbers, Wrong Comparison

Other times the money is visible, but it gets judged against the wrong alternative.

- **Gold.** People say gold "holds its value", but holding it forgoes the interest savings would earn: one reason gold prices tend to fall when interest rates rise.
- **An owner-run shop.** The books show $30,000 of profit; if the owner could earn $45,000 in a job, the shop is really losing $15,000. Accountants leave that cost out; economists put it in. Though if being your own boss is worth more than $15,000 to you, staying still passes the rule.

From One Decision to Technological Progress

Why did technological progress take off, and why does it keep going? Think of firms running the same comparison you just ran.

- A firm weighing a new machine against its old one is making the same kind of decision: benefit, direct cost, and the **opportunity cost** of the option it gives up.
- In the eighteenth century, labor was expensive and coal was cheap, so the **opportunity cost** of keeping costly workers kept growing.

From One Decision to Technological Progress

- First movers earned **innovation rents**: profit above the old way of working. Rents like that are an **incentive**: a reward that changes what an action is worth.
- The same logic runs today, as firms weigh software, robots, and AI.

But if adopting pays, everyone eventually adopts. Do the rents survive?

Wrapping Up

- Today we covered:
  - how to put a monetary number on non-monetary **costs** and **benefits**.
  - **opportunity cost**: the economic cost of an action is not just its direct cost, as in accounting, but also the forgone alternative, and that cost counts too.
- Before next class: read section **2.2**, go over the **slides**, and work the **practice problems**.
- Next class: comparative advantage, and why anyone bothers to trade.